Problem 2

Show by example that it is not always true that $\arg(ab) = \arg a + \arg b$ when using the principal value determination

$$-\pi < \arg z \leq \pi \quad (z \in C^*)$$

tocsectionProblem 2: Failure of argument addition formula for principal values

Solution

We will demonstrate that the argument addition formula can fail when using principal values.

Step 1: Understanding the Principal Argument

The **principal argument** $\arg(z)$ is defined as the unique value θ such that:

$$-\pi < \theta \leq \pi$$

and $z = |z|e^{i\theta}$.

The general argument is multi-valued: $\text{Arg}(z) = \arg(z) + 2\pi k$ for any integer k.

Step 2: The Issue with Principal Values

The formula $\text{Arg}(ab) = \text{Arg}(a) + \text{Arg}(b)$ is true for the **multi-valued** argument function (up to multiples of 2π).

However, for the **principal value**, we have:

$$\arg(ab) = \arg(a) + \arg(b) + 2\pi k$$

where k is chosen so that the result lies in $(-\pi, \pi]$.

Usually $k = 0$, but when $\arg(a) + \arg(b)$ falls outside $(-\pi, \pi]$, we need $k \neq 0$, causing the formula to fail.

Step 3: Construct a Counterexample

Let's choose two complex numbers whose principal arguments are both positive and sum to more than π.

Example:

$$a = -1 + i, \quad b = -1 + i$$

Find $\arg(a)$

$$a = -1 + i \quad \text{(point in Quadrant II)}$$

$$|a| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\tan(\text{angle from positive real axis}) = \frac{1}{-1} = -1$$

Since the point is in Quadrant II:

$$\arg(a) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

Find $\arg(b)$

$$b = -1 + i = a$$

$$\arg(b) = \frac{3\pi}{4}$$

Compute $\arg(a) + \arg(b)$

$$\arg(a) + \arg(b) = \frac{3\pi}{4} + \frac{3\pi}{4} = \frac{3\pi}{2}$$

Note that $\frac{3\pi}{2} > \pi$, so this is **outside** the principal value range $(-\pi, \pi]$.

Compute ab

$$\begin{aligned} ab = (-1+i)(-1+i) &= (-1)^2 + 2(-1)(i) + (i)^2 \\ &= 1 - 2i - 1 = -2i \end{aligned}$$

Find $\arg(ab)$

$$ab = -2i = 0 - 2i \quad \text{(point on negative imaginary axis)}$$

The point $-2i$ lies on the negative imaginary axis, so:

$$\arg(ab) = \arg(-2i) = -\frac{\pi}{2}$$

Step 4: Verify the Discrepancy

We have:

$$\arg(a) + \arg(b) = \frac{3\pi}{2}$$

$$\arg(ab) = -\frac{\pi}{2}$$

Clearly:

$$\arg(ab) \neq \arg(a) + \arg(b)$$

$$-\frac{\pi}{2} \neq \frac{3\pi}{2}$$

Step 5: Explain the Relationship

However, they differ by 2π:

$$\arg(ab) = \arg(a) + \arg(b) - 2\pi$$

$$-\frac{\pi}{2} = \frac{3\pi}{2} - 2\pi$$

This shows that:

$$\boxed{\arg(ab) = \arg(a) + \arg(b) \pmod{2\pi}}$$

but the principal values don't add directly when the sum exceeds π (or is less than $-\pi$).

Step 6: Another Example

Example 2: Let $a = -1$ and $b = i$.

$$\arg(a) = \arg(-1) = \pi$$

$$\arg(b) = \arg(i) = \frac{\pi}{2}$$

$$\arg(a) + \arg(b) = \pi + \frac{\pi}{2} = \frac{3\pi}{2}$$

$$ab = (-1)(i) = -i$$

$$\arg(ab) = \arg(-i) = -\frac{\pi}{2}$$

Again:

$$\arg(ab) = -\frac{\pi}{2} \neq \frac{3\pi}{2} = \arg(a) + \arg(b)$$

But:

$$-\frac{\pi}{2} = \frac{3\pi}{2} - 2\pi$$

General Rule

The correct formula for principal arguments is:

$$\arg(ab) = \arg(a) + \arg(b) + 2\pi k$$

where $k \in \{-1, 0, 1\}$ is chosen so that $-\pi < \arg(ab) \leq \pi$.
Specifically:

- If $\arg(a) + \arg(b) > \pi$: use $k = -1$
- If $-\pi < \arg(a) + \arg(b) \leq \pi$: use $k = 0$
- If $\arg(a) + \arg(b) \leq -\pi$: use $k = 1$

Final Answer

Counterexample:

$$\boxed{a = -1 + i, \quad b = -1 + i}$$

Results:

$$\boxed{\begin{aligned} \arg(a) &= \frac{3\pi}{4} \\ \arg(b) &= \frac{3\pi}{4} \\ \arg(a) + \arg(b) &= \frac{3\pi}{2} \\ ab &= -2i \\ \arg(ab) &= -\frac{\pi}{2} \\ \arg(ab) &\neq \arg(a) + \arg(b) \end{aligned}}$$

Conclusion: The formula $\arg(ab) = \arg(a) + \arg(b)$ fails for principal values when the sum falls outside $(-\pi, \pi]$. The correct relationship is:

$$\boxed{\arg(ab) \equiv \arg(a) + \arg(b) \pmod{2\pi}}$$

with an appropriate adjustment of $\pm 2\pi$ to bring the result into the principal range.